Ethanol Blending in Gasoline – Dominican Republic

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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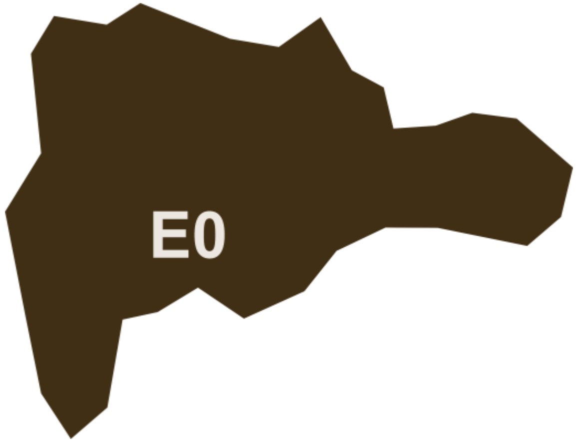
Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

Ethanol Blending in Gasoline – Dominican Republic

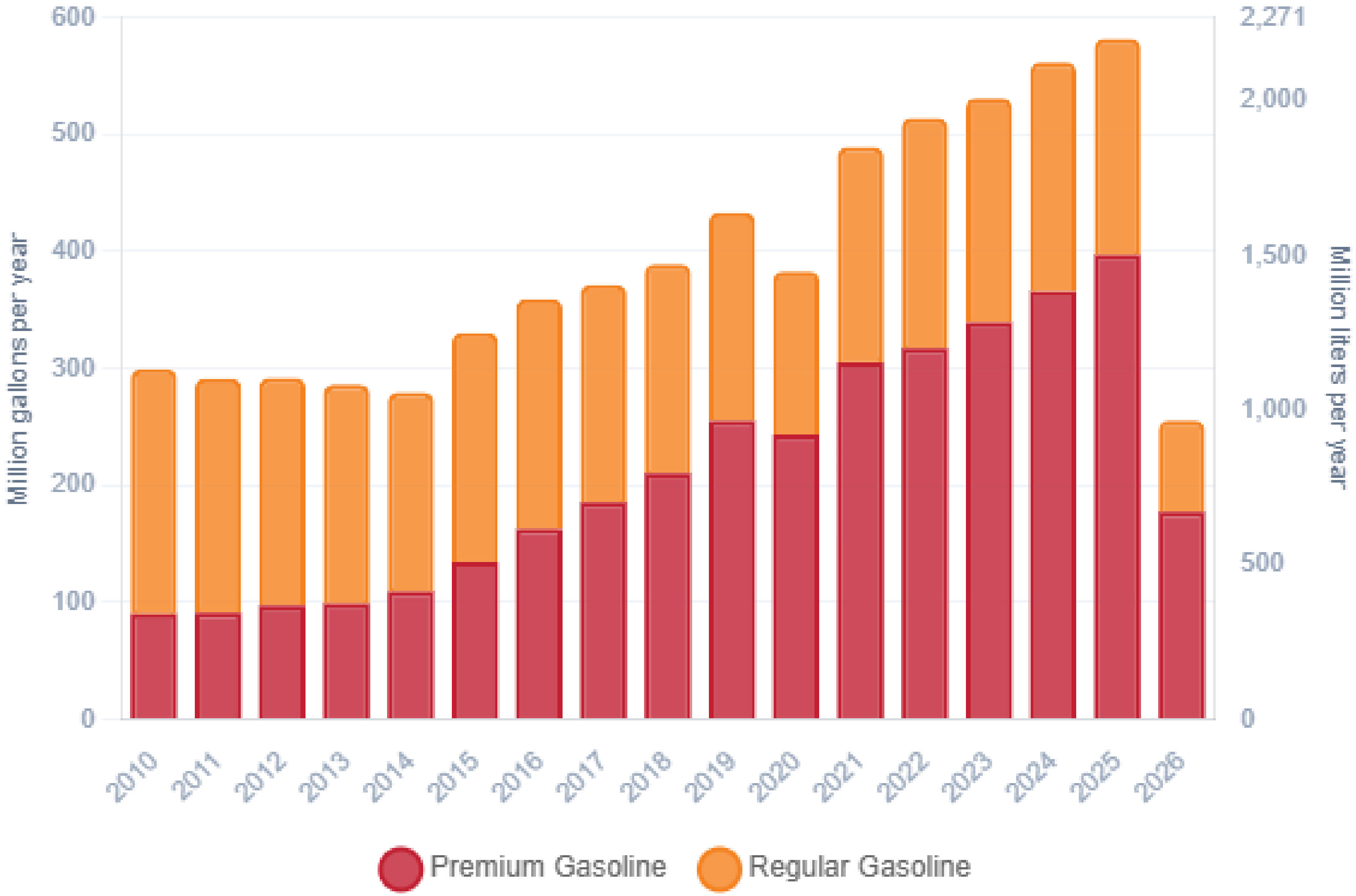


National demand is covered up by local production representing 12%, and imports from United States, Trinidad and Tobago, Caribbean, among other countries. In 2025, gasoline consumption was 581 million gallons (2,200 million liters). Regular gasoline consumption was 32% of total demand and premium 68%. The Dominican Government applies subsidies to gasoline prices.

Dominican Republic does not currently have an ethanol mandate.

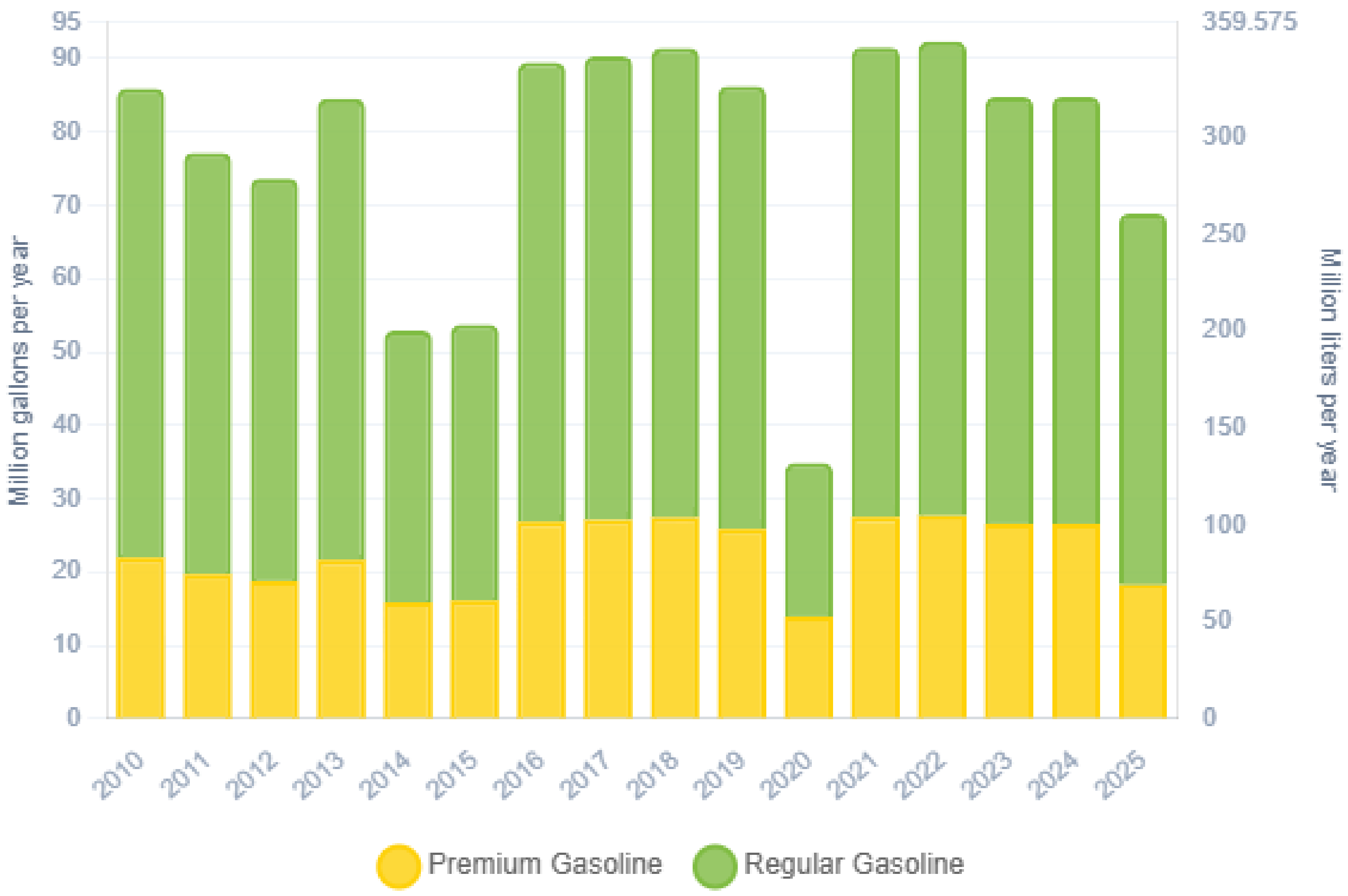
Source: DGII

Gasoline Demand in Dominican Republic



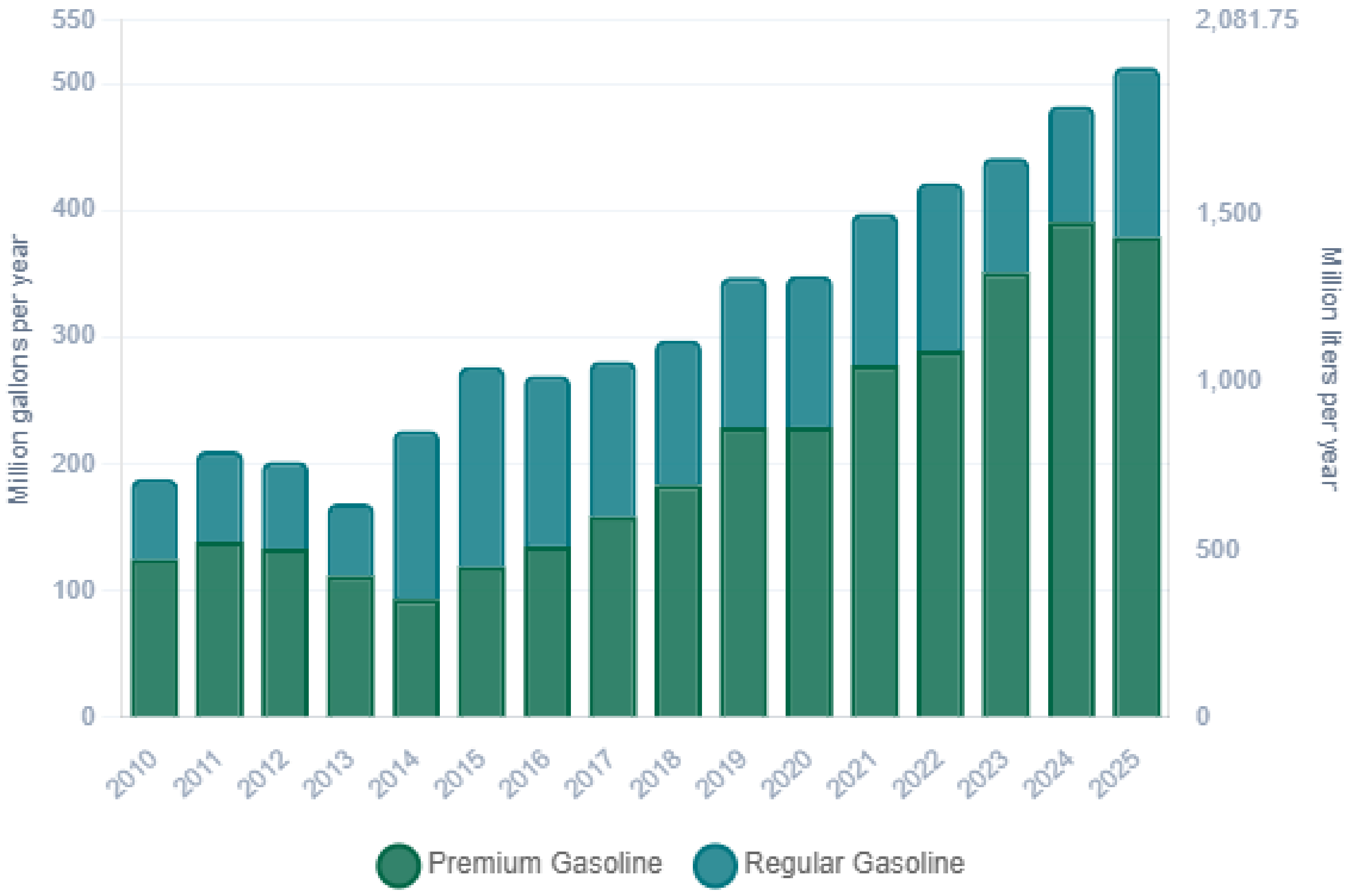
Source:
DGII

Gasoline Production in Dominican Republic



Source:
DGII

Gasoline Imports in Dominican Republic



Source:
DGII

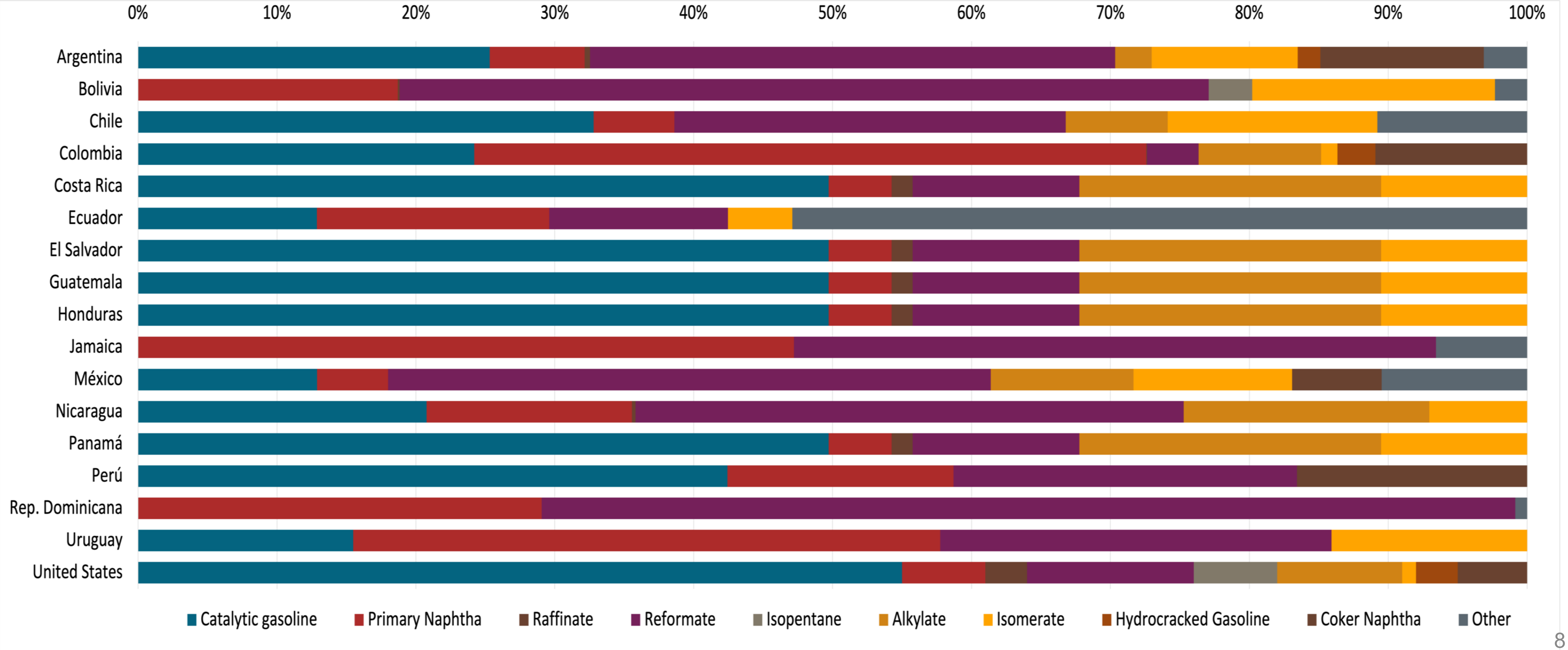
Gasoline Quality in Dominican Republic

Name	NORDOM 476 3rd Revision				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2015				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Regular Gasoline	Premium Gasoline	Oxygenated Regular Gasoline	Oxygenated Premium Gasoline	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	-	-	-	-	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	-	-	-	-	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 89	> 95	> 90	> 96	> 95	> 95	> 98	> 98
MON	> 76	> 82	> 77	> 83	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3.5 %m/m	< 3.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)			< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 61 kPa	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: NORDOM

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



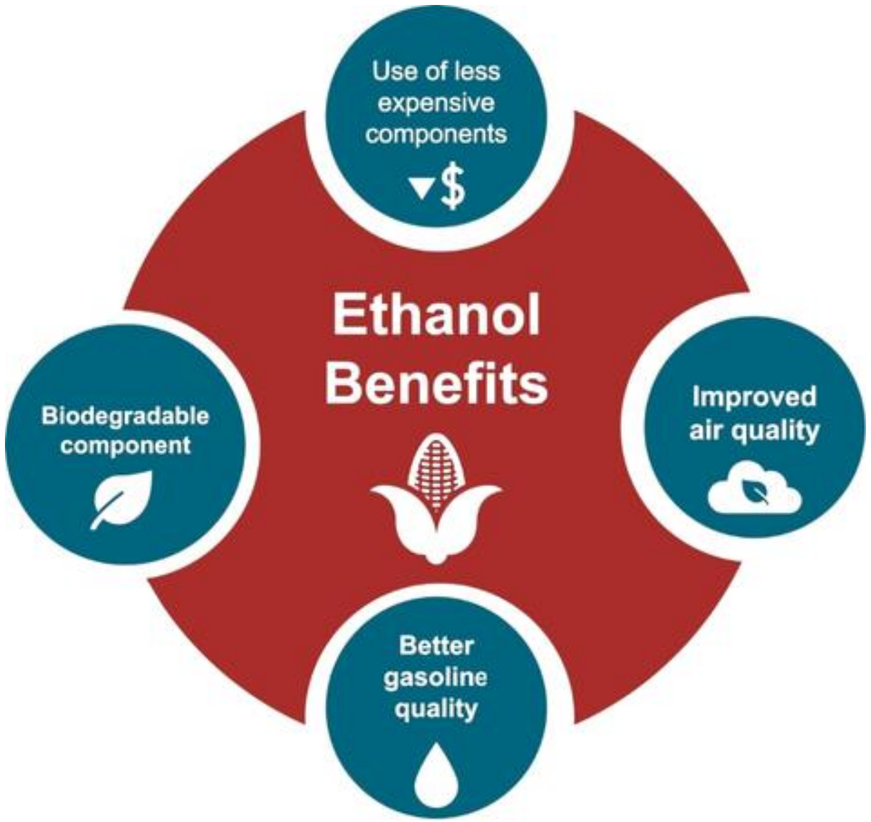
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

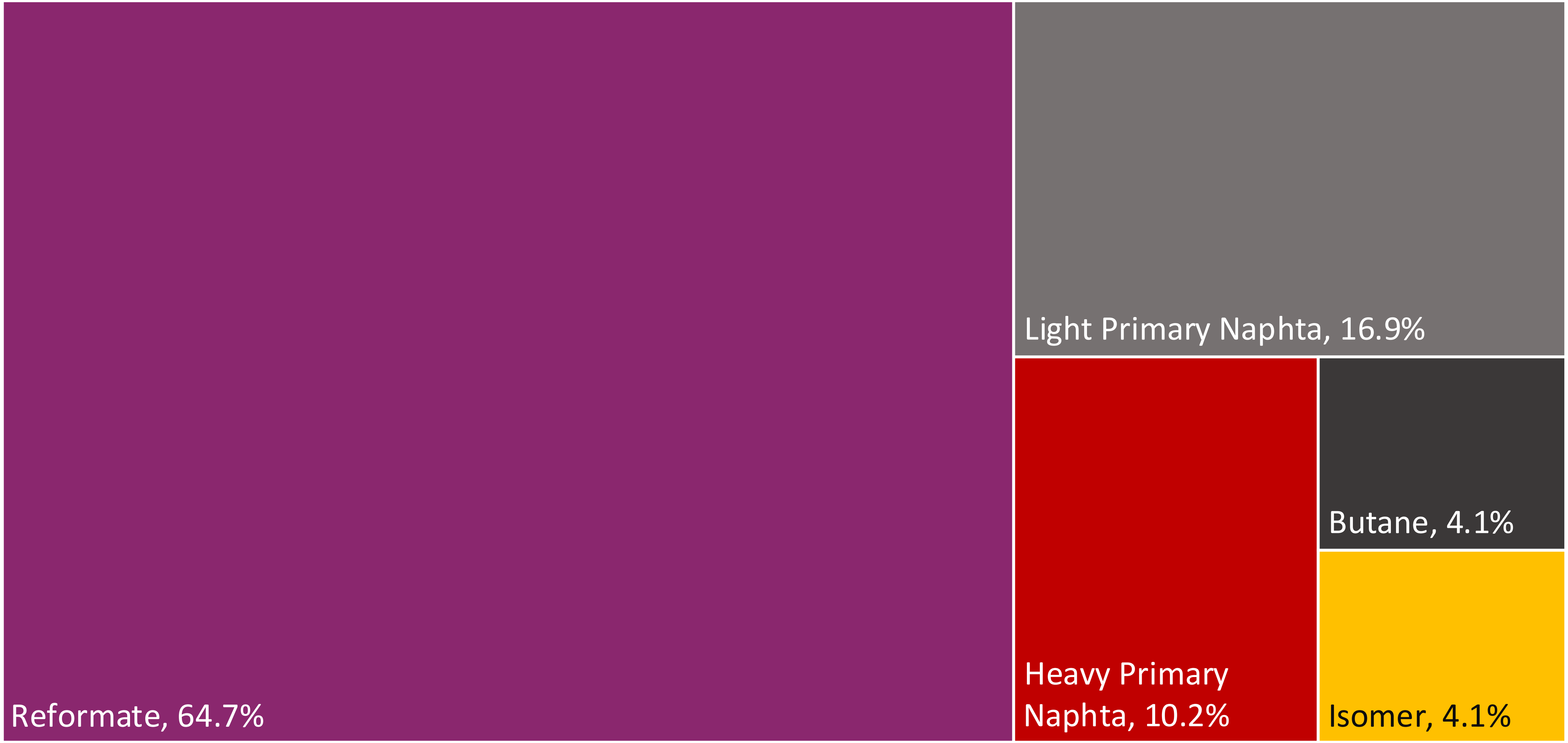
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

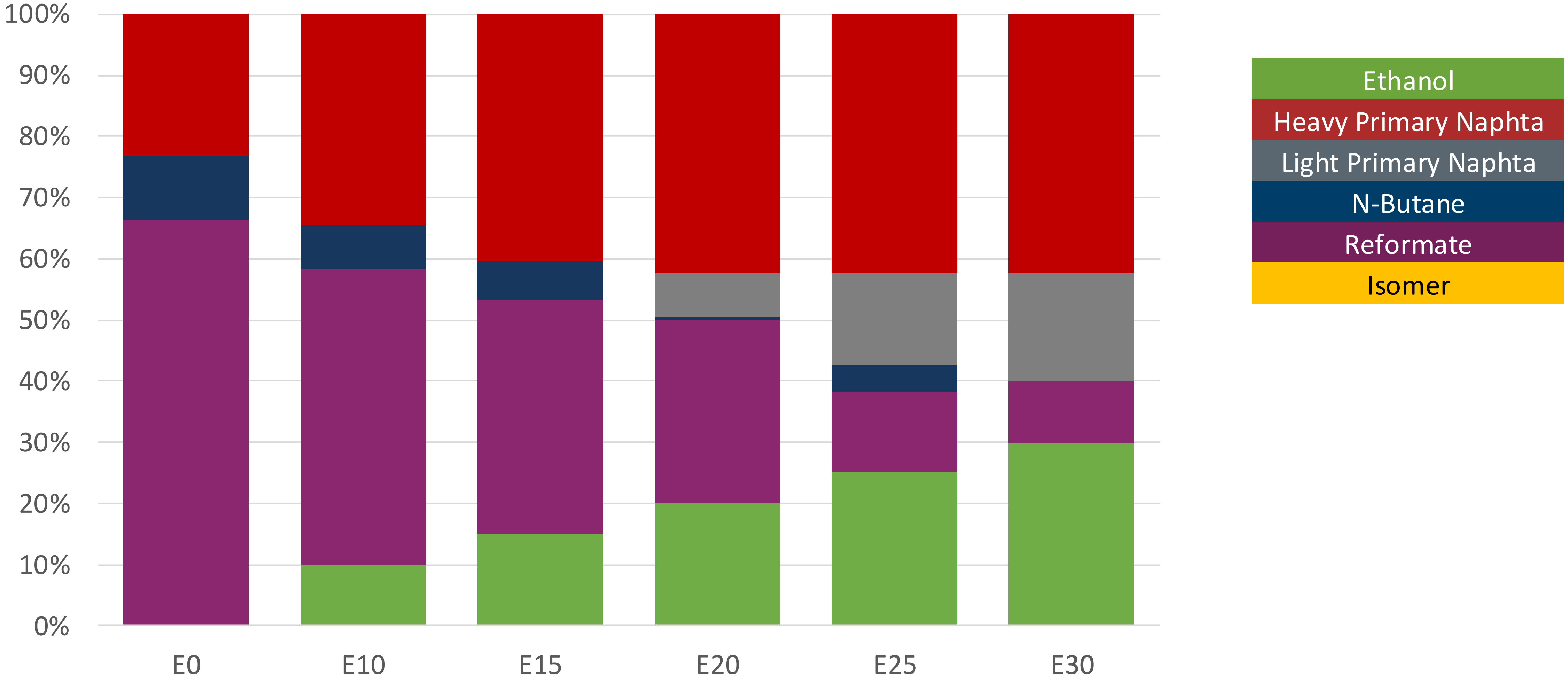


Available Blending Components



Source: Faro90

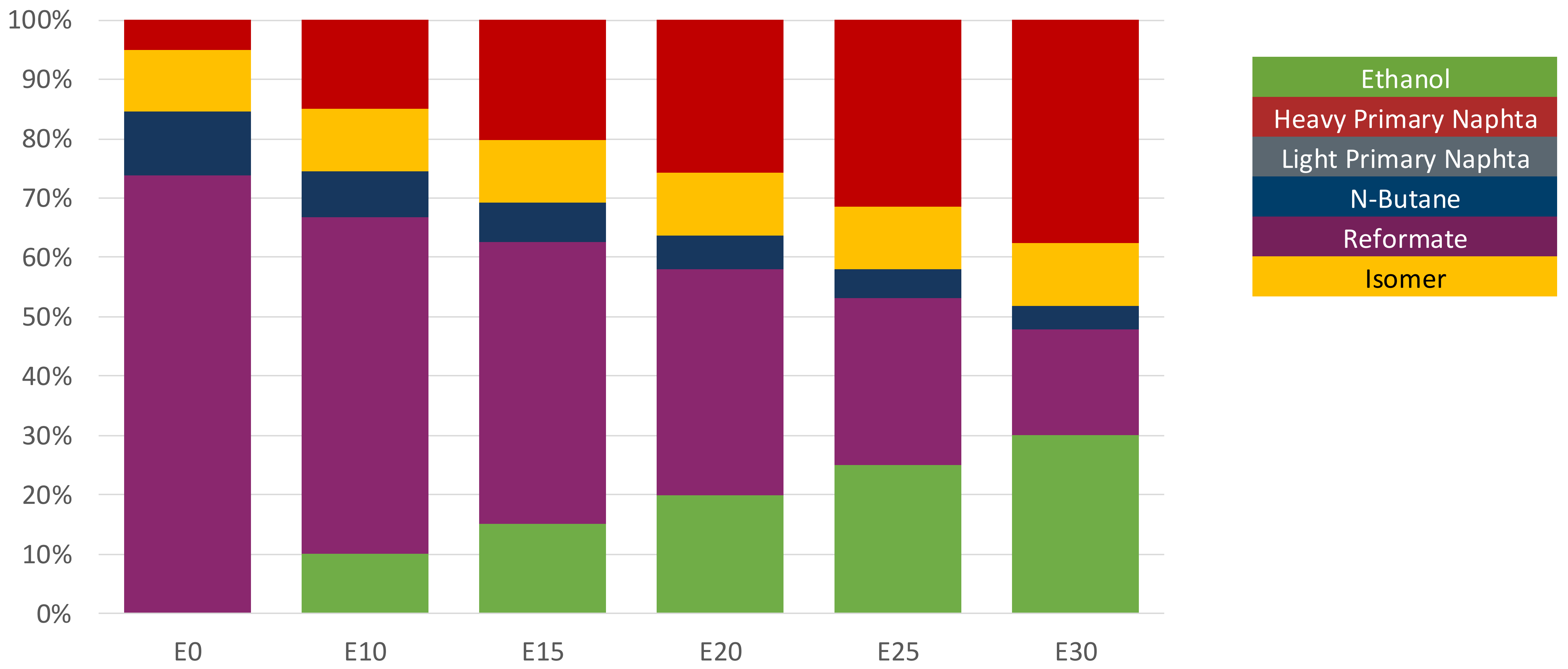
Dominican Republic – Regular – Constante Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	90.0	90.0	90.0	90.0	90.0	91.8
Price (US$/gal)	$ 2.146	$ 1.849	$ 1.684	$ 1.589	$ 1.338	$ 1.328

Source: Faro90

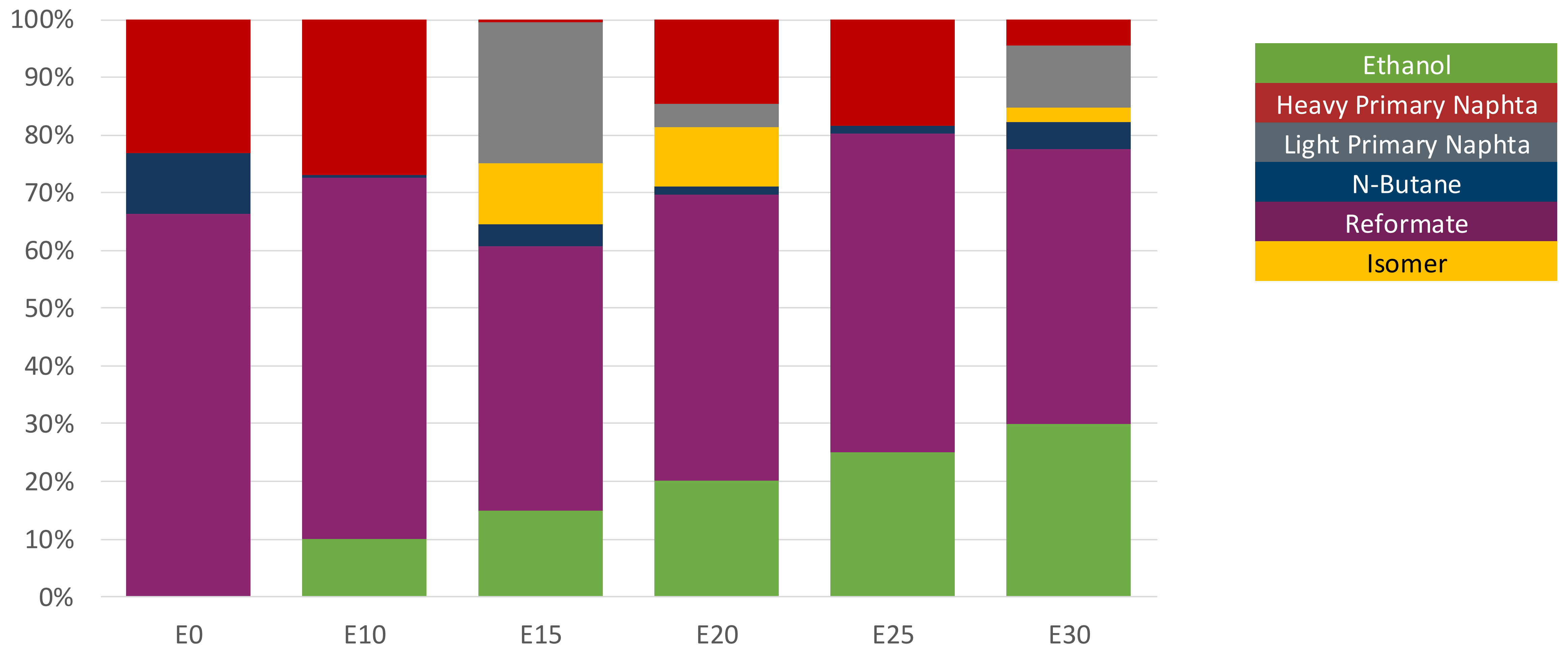
Dominican Republic– Premium – Constant Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	96.0	96.0	96.0	96.0	96.0	96.0
Price (US$/gal)	$ 2.454	$ 2.184	$ 2.035	$ 1.877	$ 1.712	$ 1.538

Source: Faro90

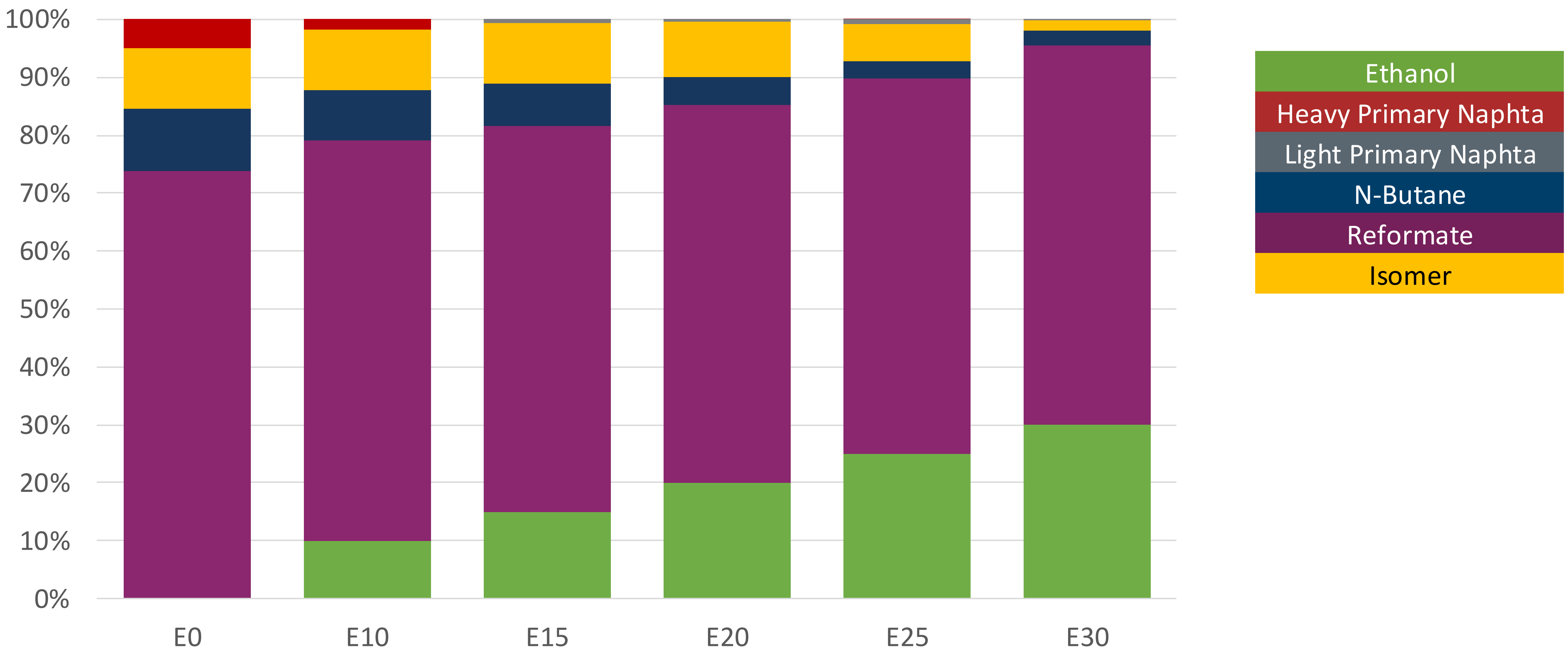
Dominican Republic – Regular – Increased Octane



Octane (RON)	90.0	93.2	96.8	99.0	101.3	104.3
Price (US$/gal)	$ 2.146	$ 2.146	$ 2.146	$ 2.146	$ 2.146	$ 2.146

Source: Faro90

Dominican Republic – Premium – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	96.0	100.8	102.9	104.6	106.3	108.2
Price (US$/gal)	$ 2.454	$ 2.454	$ 2.454	$ 2.454	$ 2.454	$ 2.454

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

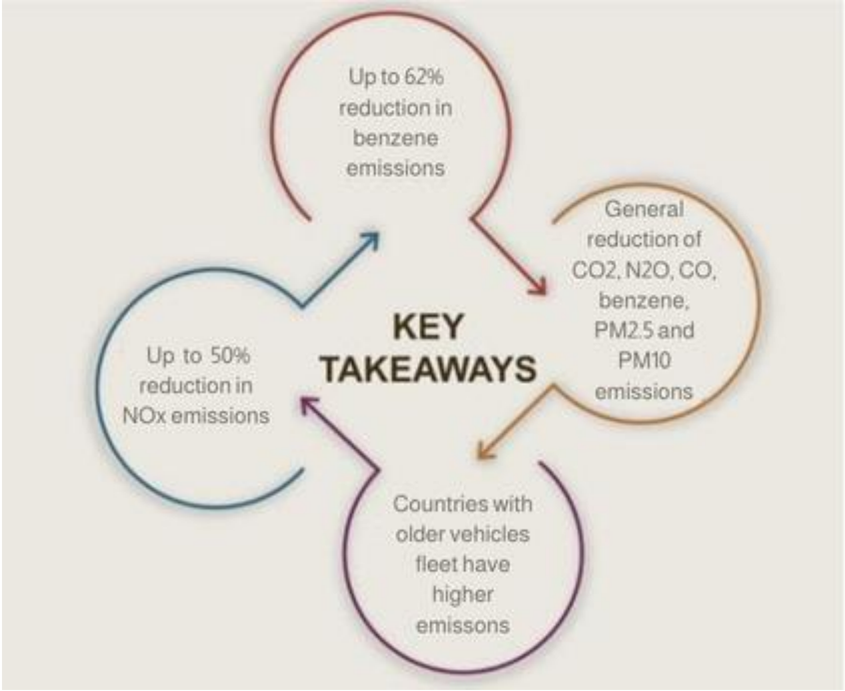
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicular Fleet in Dominican Republic

FARO90 HCX

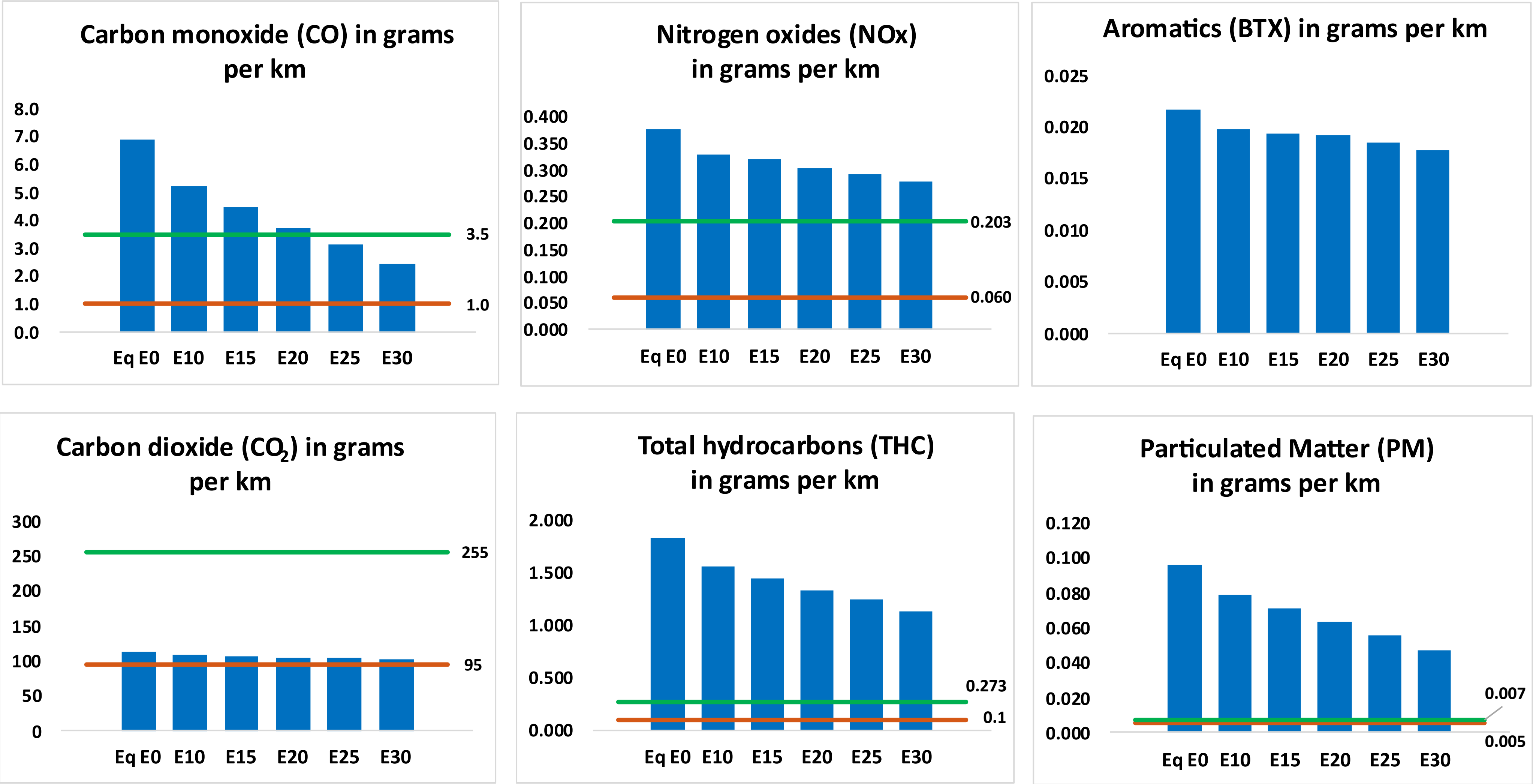
Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	102,256
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	153,384
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	291,768
		Euro III		> 8 years, > 160 tkm	102,402
		Euro IV		> 8 years, > 160 tkm	75,866
		Euro V / Euro 6		> 8 years, > 160 tkm	107,925
				4-8 years, 80-160 tkm	178,225
				< 4 years, < 80 tkm	148,308
	Hybrid			Any	7,337
	Electric			Any	5,510
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	27,660
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	41,490
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	119,696
		Euro III		> 8 years, > 160 tkm	195,723
		Euro IV		> 8 years, > 160 tkm	133,309
		Euro V / Euro 6		> 8 years, > 160 tkm	126,253
				4-8 years, 80-160 tkm	213,698
				< 4 years, < 80 tkm	330,755
	Hybrid			Any	7,400
	Electric			Any	5,670
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	839,636
		Euro II		> 8 years, > 160 tkm	86,144
		Euro III		> 8 years, > 160 tkm	1,182,587
		Euro IV		> 8 years, > 160 tkm	164,270
				4-8 years, 80-160 tkm	519,609
		Euro V		4-8 years, 80-160 tkm	133,556
				< 4 years, < 80 tkm	900,975
	Electric			Any	19,916
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	36,882
	FI	3-Ways		> 8 years, > 160 tkm	172,522
		Euro III		> 8 years, > 160 tkm	50,357
		Euro IV		> 8 years, > 160 tkm	34,363
		Euro V / Euro 6		> 8 years, > 160 tkm	28,094
				4-8 years, 80-160 tkm	48,287
				< 4 years, < 80 tkm	73,979
	Hybrid			Any	282
	Electric			Any	976

Gasoline Vehicular Fleet: **5,455,680**
Average Age: **10 years**
Motorcycles: **62%**

Source: Dirección General de Impuestos Internos (DGII), 2024

Vehicular Emissions – Dominican Republic

United States
Euro 6 Standard



Source: Faro90

Dominican Republic – Vehicular Fleet

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	941,919	828,510	715,101	614,167	508,913	426,867	336,644	-24%	-46%
CO2	15,484,691	15,086,513	14,710,456	14,414,699	14,268,346	14,209,005	13,947,092	-5%	-8%
VOC	248,823	230,217	211,611	196,467	181,273	169,604	153,982	-15%	-27%
NOx	51,415	47,473	44,844	43,545	41,407	39,752	37,790	-13%	-19%
SOx	171	158	145	134	124	112	100	-15%	-28%
NH3	8,851	8,763	8,676	8,717	8,815	8,884	8,941	-2%	0%
N2O	195	194	193	194	198	200	202	-1%	2%
CH4	55,203	55,203	55,203	56,031	57,245	58,294	59,288	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,204	1,102	1,000	915	829	762	674	-17%	-31%
Acetaldehyde	4,414	5,066	7,433	11,320	15,422	17,623	20,823	68%	249%
Formaldehyde	18,492	17,978	20,957	24,608	25,781	28,520	31,048	13%	39%
Benzene	2,959	2,843	2,694	2,651	2,629	2,515	2,433	-9%	-11%
PM2.5	1,101	1,000	900	812	726	634	537	-18%	-34%
PM10	12,004	10,913	9,822	8,861	7,922	6,914	5,860	-18%	-34%

Source: Faro90

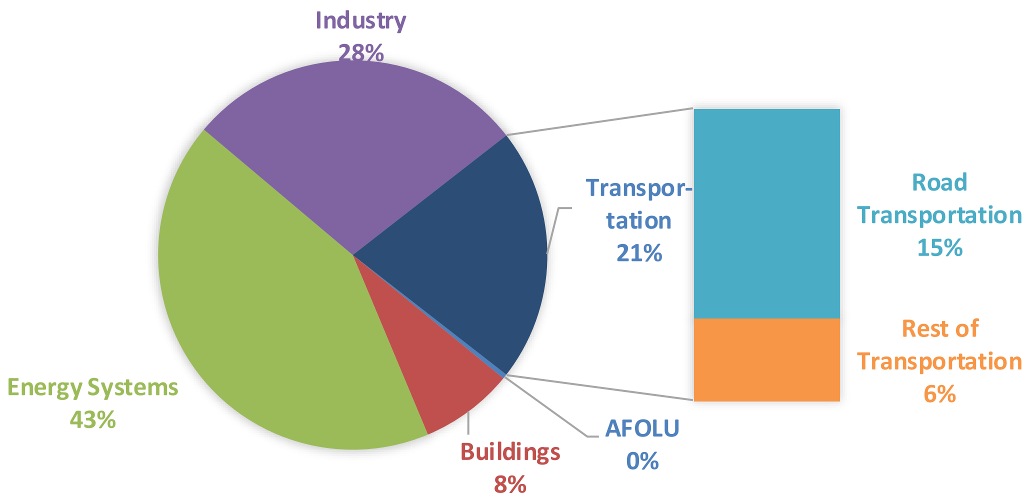
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

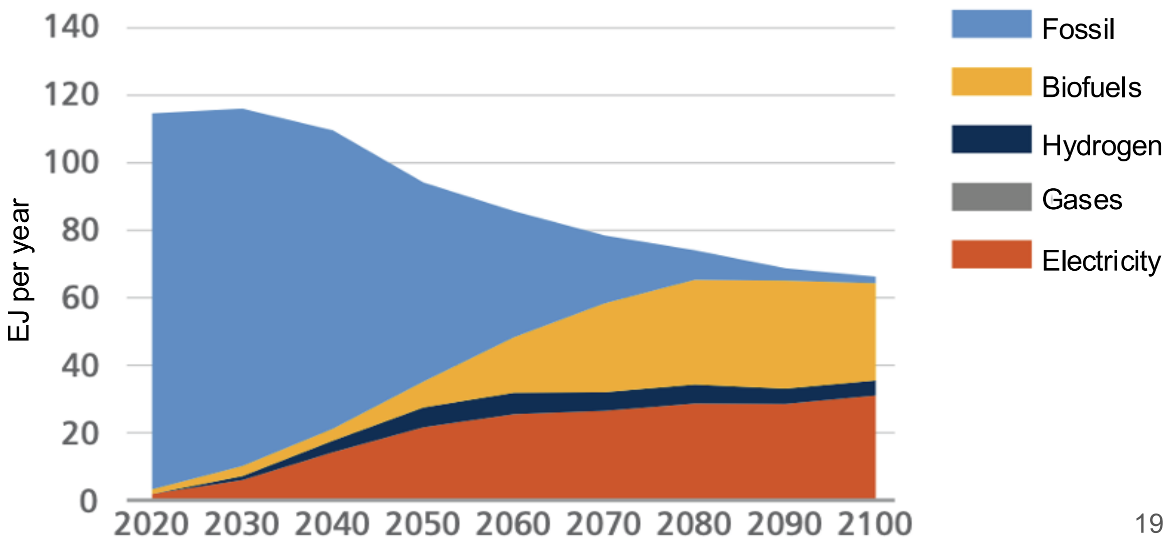
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

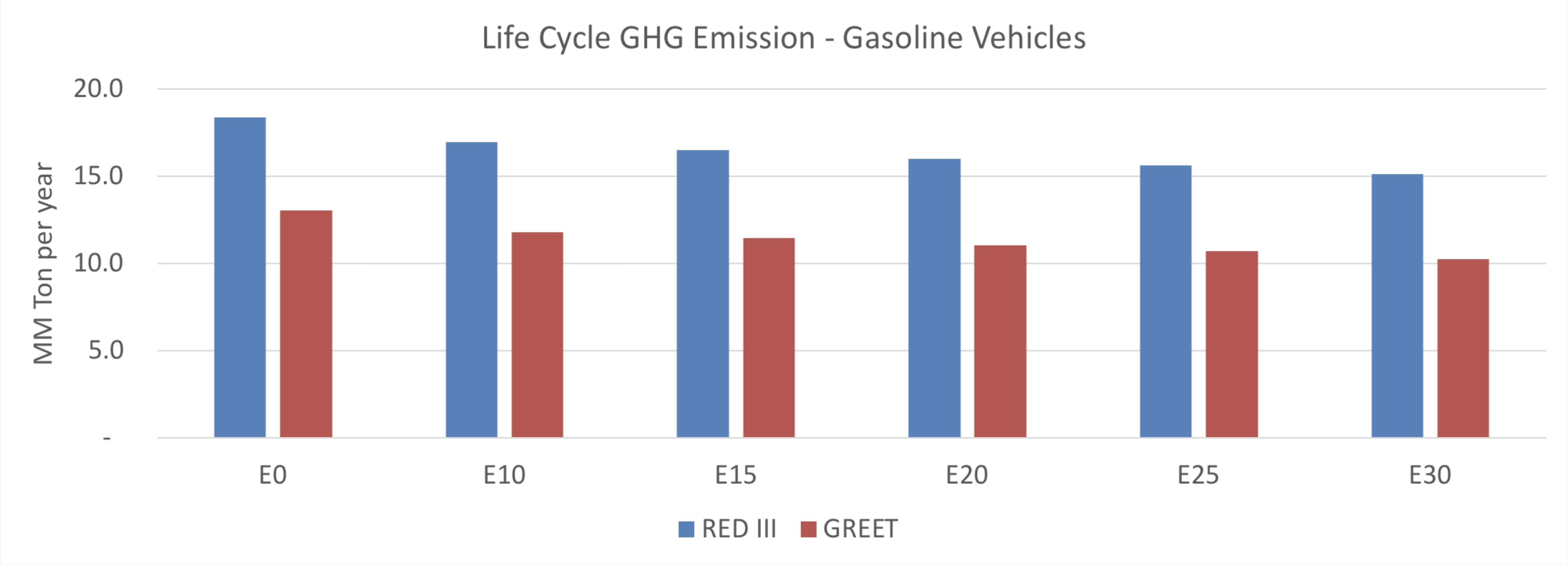
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Dominican Republic – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	51.6
2035 Target (MMT/year)	36.1
Delta	15.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.5%	12.4%	15.5%	18.1%	21.5%
RED II/III	0.0%	7.7%	10.2%	12.8%	15.0%	17.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	8.0%	10.4%	13.1%	15.2%	18.1%
RED II/III	0.0%	9.2%	12.1%	15.2%	17.8%	21.1%

Source: Greet, RED II/III, climateactiontracker.org, F90